White-Light YOLO Audit v1: Results and Evaluation Architecture

Forensic Bruise Segmentation Pipeline

Generated from `wl_audit_v1` outputs

Scope

This report summarizes only the `yolo_wl_audit_v1.py` diagnostic result for the white-light YOLO model audit. The evaluated model in the provided output is `yolo_sem_distilled`. The script compares three inference/evaluation paths:

1. **Custom PyTorch path at 640, no temperature scaling**: raw YOLO module output, validation-selected threshold at $T = 1.0$.
2. **Custom PyTorch path at 640, temperature scaled**: raw YOLO module output, validation-selected temperature and threshold.
3. **Ultralytics native path**: `YOLO.predict()` with native Ultralytics postprocessing and native-resolution mask comparison.

Result Summary

Table 1: Main test-set segmentation metrics, $n = 185$ white-light test images.

Metric	640 No Temp	640 Temp Scaled	Ultralytics Native
Mean Dice	0.6072	0.6080	**0.7412**
Median Dice	0.7246	0.7264	**0.8426**
Mean IoU	0.4991	0.4987	**0.6397**
Median IoU	0.5681	0.5704	**0.7279**
Mean Precision	0.7347	0.7261	**0.8754**
Median Precision	0.7967	0.7856	**0.9316**
Mean Recall	0.6445	0.6542	**0.7258**
Median Recall	0.7751	0.7996	**0.8131**
Mean Pred/GT Ratio	0.9893	1.0259	0.8724
Median Pred/GT Ratio	0.9711	0.9989	0.9024

Table 2: Failure metrics on the same 185-image test set.

Metric	640 No Temp	640 Temp Scaled	Ultralytics Native
Zero Dice Count	24 / 185	23 / 185	**13 / 185**
Zero Dice Rate	12.97%	12.43%	**7.03%**
Complete Miss Count	19 / 185	18 / 185	**13 / 185**
Complete Miss Rate	10.27%	9.73%	**7.03%**

Table 3: Validation-selected threshold/temperature and 640-only speed.

Setting / Speed Metric	Value
Validation best temperature	6.00
Validation best threshold	0.40
No-temperature threshold at $T = 1.0$	0.10
Raw forward FPS at 640	167.66 FPS
Raw forward mean time	5.96 ms
Raw forward std. time	0.04 ms
Full 640 pipeline FPS	51.50 FPS
Full 640 pipeline mean time	19.42 ms
Full 640 pipeline std. time	9.97 ms

Architecture / Evaluation Paths

A. Custom PyTorch Probability-Map Path at 640

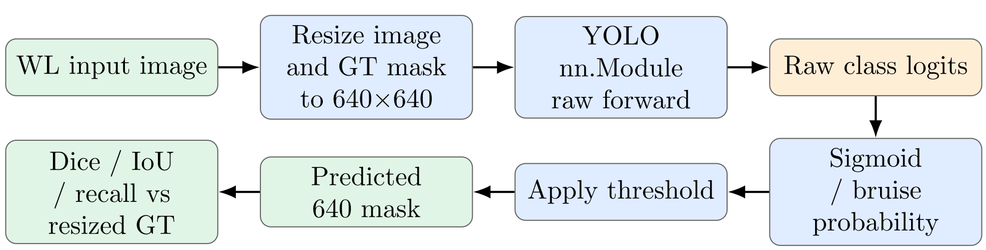

Figure 1: Custom path used for both no-temperature and temperature-scaled 640 evaluations.

No-temperature branch: $T = 1.0$, threshold $= 0.10$.
Temperature-scaled branch: validation-selected $T = 6.0$, threshold $= 0.40$.

B. Ultralytics Native Path

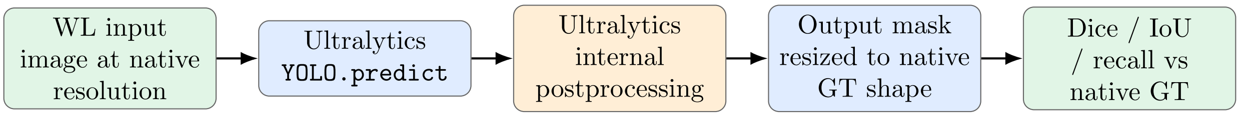

Figure 2: Native Ultralytics evaluation path. This path differs from the 640 custom path because final comparison is at native image/mask resolution.

C. FPS Timing Definition

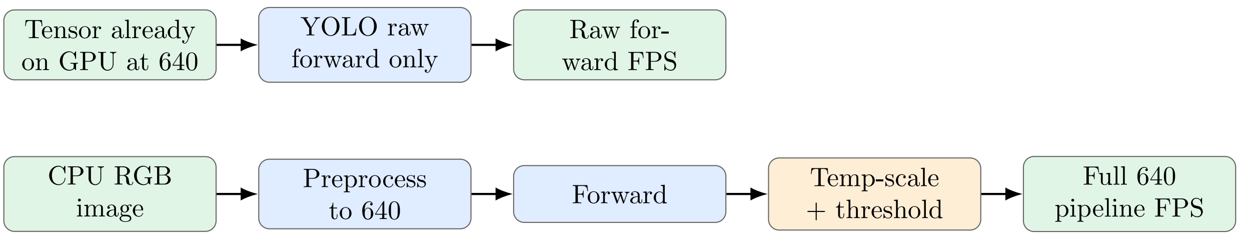

Figure 3: FPS definitions. The audit intentionally does not include resizing the final mask back to 6024×4022 or other camera-native resolution in the timed 640 pipeline.

Interpretation

- **Ultralytics native evaluation is clearly stronger** in this audit: mean Dice 0.7412 and median Dice 0.8426, compared with roughly 0.608 mean Dice and 0.725–0.726 median Dice for the custom 640 probability-map path.
- **Temperature scaling gives only a marginal improvement**: mean Dice increases from 0.6072 to 0.6080, and complete miss rate decreases from 10.27% to 9.73%.
- **The main gap is not explained by temperature scaling**. The larger difference is between the custom 640 evaluation route and the Ultralytics native route, suggesting that preprocessing/postprocessing alignment is the next debugging target.
- **Speed at 640 is high for raw forward** at 167.66 FPS, but full 640 pipeline throughput is lower at 51.50 FPS because preprocessing and postprocessing are included.

Result Files Used

- `audit_summary.csv`
- `test_per_image_640_no_temp.csv`
- `test_per_image_640_temp_scaled.csv`
- `test_per_image_ultralytics_native.csv`
- Script reference: `scripts/yolo_wl_audit_v1.py`